

AMANI KIRUGA

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EDUCATION

PhD in Computer Science, Harvard University Expected 2030
Advisor: Prof. Yilun Du | Research Focus: Scene understanding for vision and robotics

B.S in Computer Science, University of Delaware August 2024
Cumulative GPA: 3.94 | Major GPA: 4.0

Selected Coursework: Machine Learning, Computer Vision, Calculus, Linear Algebra, Probability Theory, Computational Mathematics, Parallel Computing, Algorithms, Data Structures, Data Science

Bachelor Thesis: "Modelling 3D Causal Processes from 2D Vision Data: Challenges and Solutions"

SKILLS

Languages	Python, C++, C, Java, JavaScript, TypeScript, HTML, CSS
Frameworks/Libraries	Pytorch, OpenCV, NumPy, Pandas, Scikit-Learn, Node.js, React.js

EXPERIENCE

Student Researcher Sep. 2023 - March 2025
MIT Computer Science and Artificial Intelligence Laboratory (CSAIL) *Cambridge, MA*

- Worked with Prof. Vincent Sitzmann and Prof. Ayush Tewari on unsupervised learning.
- Developed a method for out-of-domain generalization for single-image 3d reconstruction.

Student Researcher June 2023 - Sep. 2023
MIT Media Lab *Cambridge, MA*

- Worked with Dr. Andrew Lippman on content analysis in the Viral Communications Group.
- Designed and built an interactive portal to analyze media sources with LLMs using Next.js and MongoDB.

Student Researcher Jan. 2024 - May 2024
University of Delaware, Dynamic Vision Lab *Newark, DE*

- Worked with Prof. Christopher Rasmussen on video action recognition and pose detection.
- Benchmarked LLM performance, trained ResNet and DINO baselines using PyTorch.

Fullstack Software Engineer Sep. 2021 - May 2024
University of Delaware, PAROT Group *Newark, DE*

- Designed and built two releases of the National Science Foundation (NSF) funded Atom research portal that provides the scientific community with accessible atomic data.
- Utilized React, Node.js, Next.js, and PostgreSQL for robust application architecture.

Student Researcher June 2021 - May 2023
University of Delaware, D-REAL Lab *Newark, DE*

- Collected and annotated a human biomechanics dataset from YouTube with 1000 videos.
- Trained a video classification model with PyTorch to detect falls in an unconstrained environment.

RESEARCH PRESENTATIONS & PUBLICATIONS

- **Robust Monocular 3D Reconstruction by Integrating Feed-forward Inference and Analysis-by-Synthesis** **A. Kiruga**, M. Hamilton, JB. Tenenbaum, WT. Freeman, V. Sitzmann, Y. Du, A. Tewari. To be submitted.
- **Infant Video Interaction Recognition using Monocular Depth Estimation** C. Rasmussen, **A. Kiruga**, J. Orlando and M. Lobo. Poster, ISVC 2024 & APTA CSM 2024.
- **A Portal for High-Precision Atomic Data and Computation** **A. Kiruga**, C. Cheung, P. Barakshan, A. Bhosale, R. Eigenmann, M. Safronova, B. Arora. Computer Physics Communications 2025.
- **Non-Hierarchical Transformers for Pedestrian Segmentation** **A. Kiruga**, X. Peng. Poster, CVPRW 2023.
- **Human Falls in The Wild: A Video Dataset** Z. Jia, **A. Kiruga**, X. Peng. Poster, UD Undergraduate Research Summer Symposium 2022.
- **Near Real-time Deep Group Profiling in Crowded Scenes** **A. Kiruga**, M. Baksh, K. Barner. Poster, UD Undergraduate Research Summer Symposium 2021.

PROJECTS

- **EmojiGAN** (Jan. 2022) Used PyTorch to train a Deep Convolutional Generative Adversarial Network (DCGAN) on an open-source emoji dataset of 4000+ pictures. Tested for GAN memorization using K-Nearest Neighbor search.

HONORS & AWARDS

- **MIT Summer Research Program Scholar** (June 2023 - Sep. 2023)
- **Google CS Research Mentorship Program Scholar** (Jan. 2022 - April 2022)
- **Eugene du Pont Memorial Scholarship** (2020-2024) - Full-ride scholarship awarded to 1 of 30,000 applicants based on high academic achievement, service, and leadership.
- **Outstanding Senior Student Award** (2024) - Awarded to 1 of 600 senior undergraduates in the CIS department.
- **Outstanding Sophomore Student/Paul D. Amer Meritorious Award** (2022) - Awarded to 2 of 600 sophomore undergraduates in the CIS department.
- **Rees Carl J & Eleanor K Mathematics Scholarship** (2022) - Awarded to 10 of 200 undergraduates based on high academic achievement in Mathematical Sciences.
- **Dean's List** (2021, 2022, 2023, 2024)

LEADERSHIP & SERVICE

Founder

University of Delaware, Competitive Programming Club

Feb. 2022 - Sep. 2024

Newark, DE

- Founded the UD Competitive programming club on campus and grew club to 100+ members.
- Organized weekly presentations and competitions to spread competitive programming knowledge.
- Received bronze medal in the Mid-Atlantic regional 2021 International Collegiate Programming Contest.

Teaching Assistant

Microsoft Philanthropies TEALS, Trenton Central High School

Sep. 2021 - May 2023

Trenton, NJ

- Assisted in the running of a class of 16 students in an AP Computer Science A classroom.
- Supported students to learn content related to Java by holding one-on-one help sessions and giving presentations.